

Financial Portfolio Analysis

Problem Statement

Build a complete financial portfolio analysis system that evaluates stock performance, measures portfolio risk, compares benchmark returns, and visualizes insights through an interactive dashboard.

This project demonstrates:

- Financial analytics
- Data cleaning and transformation
- SQL and Python skills
- Data visualization
- Business storytelling
- Dashboard creation

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib / Plotly
- SQL
- Power BI / Tableau
- Yahoo Finance API (yfinance)
- Jupyter Notebook

Business Problem

An investor wants to:

- Track stock portfolio performance
- Understand portfolio risk
- Compare returns with benchmark indices
- Identify best and worst-performing assets
- Make data-driven investment decisions

The project provides analytical insights into:

- Returns
- Volatility
- Portfolio allocation
- Risk-adjusted performance

Importing Libraries

```
In [7]: import yfinance as yf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Download Stock Data

```
In [8]: stocks = ['RELIANCE.NS', 'TCS.NS', 'INFY.NS', 'HDFCBANK.NS']

start_date = '2023-01-01'
end_date = '2025-01-01'

stock_data = yf.download(stocks, start=start_date, end=end_date)['Close']

print(stock_data.head())
```

[*****100%*****] 4 of 4 completed

Ticker	HDFCBANK.NS	INFY.NS	RELIANCE.NS	TCS.NS
Date				
2023-01-02	783.677856	1397.702148	1175.889648	2932.793701
2023-01-03	788.802368	1396.372314	1167.284668	2977.666016
2023-01-04	774.704163	1370.921997	1149.709595	2980.633057
2023-01-05	769.724060	1352.946411	1147.655273	2977.440918
2023-01-06	767.173889	1328.459106	1158.086304	2887.922363

Calculate Daily Returns

```
In [9]: daily_returns = stock_data.pct_change()

print(daily_returns.head())
```

Ticker	HDFCBANK.NS	INFY.NS	RELIANCE.NS	TCS.NS
Date				
2023-01-02	NaN	NaN	NaN	NaN
2023-01-03	0.006539	-0.000951	-0.007318	0.015300
2023-01-04	-0.017873	-0.018226	-0.015056	0.000996
2023-01-05	-0.006428	-0.013112	-0.001787	-0.001071
2023-01-06	-0.003313	-0.018099	0.009089	-0.030066

Portfolio Allocation

Assume:

Stock Allocation

- Reliance - 30%
- TCS - 25%
- Infosys - 25%
- HDFC Bank - 20%

```
In [14]: weights = np.array([0.30, 0.25, 0.25, 0.20])
```

Portfolio Returns

```
In [15]: portfolio_returns = daily_returns.dot(weights)

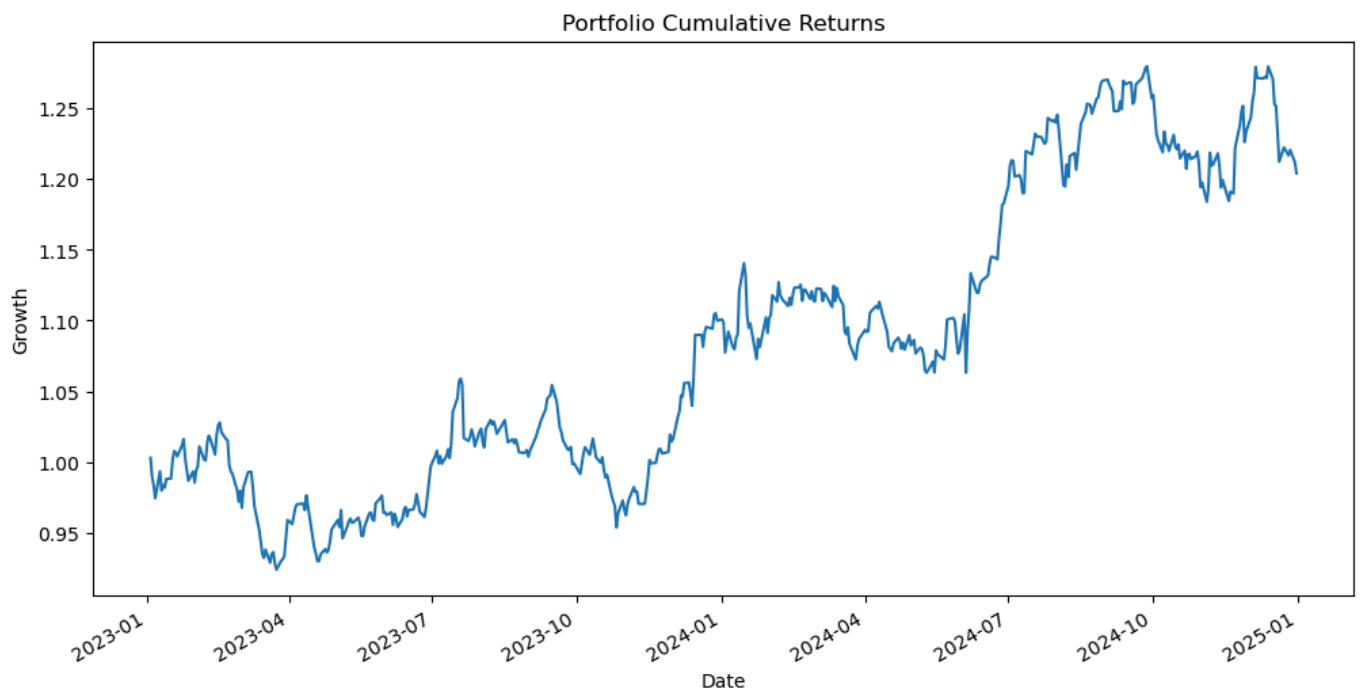
print(portfolio_returns.head())
```

```
Date
2023-01-02      NaN
2023-01-03    0.002954
2023-01-04   -0.013483
2023-01-05   -0.005867
2023-01-06   -0.009260
dtype: float64
```

Cumulative Returns

```
In [16]: cumulative_returns = (1 + portfolio_returns).cumprod()

cumulative_returns.plot(figsize=(12,6))
plt.title('Portfolio Cumulative Returns')
plt.xlabel('Date')
plt.ylabel('Growth')
plt.show()
```



Portfolio Metrics

Annualized Return

```
In [17]: annual_return = portfolio_returns.mean() * 252
print('Annual Return:', annual_return)
```

Annual Return: 0.10563056794399583

Annualized Volatility

```
In [18]: annual_volatility = portfolio_returns.std() * np.sqrt(252)
print('Annual Volatility:', annual_volatility)
```

Annual Volatility: 0.14275330855329596

Sharpe Ratio

```
In [19]: risk_free_rate = 0.06

         sharpe_ratio = (annual_return - risk_free_rate) / annual_volatility

         print('Sharpe Ratio:', sharpe_ratio)
```

Sharpe Ratio: 0.31964630737059224

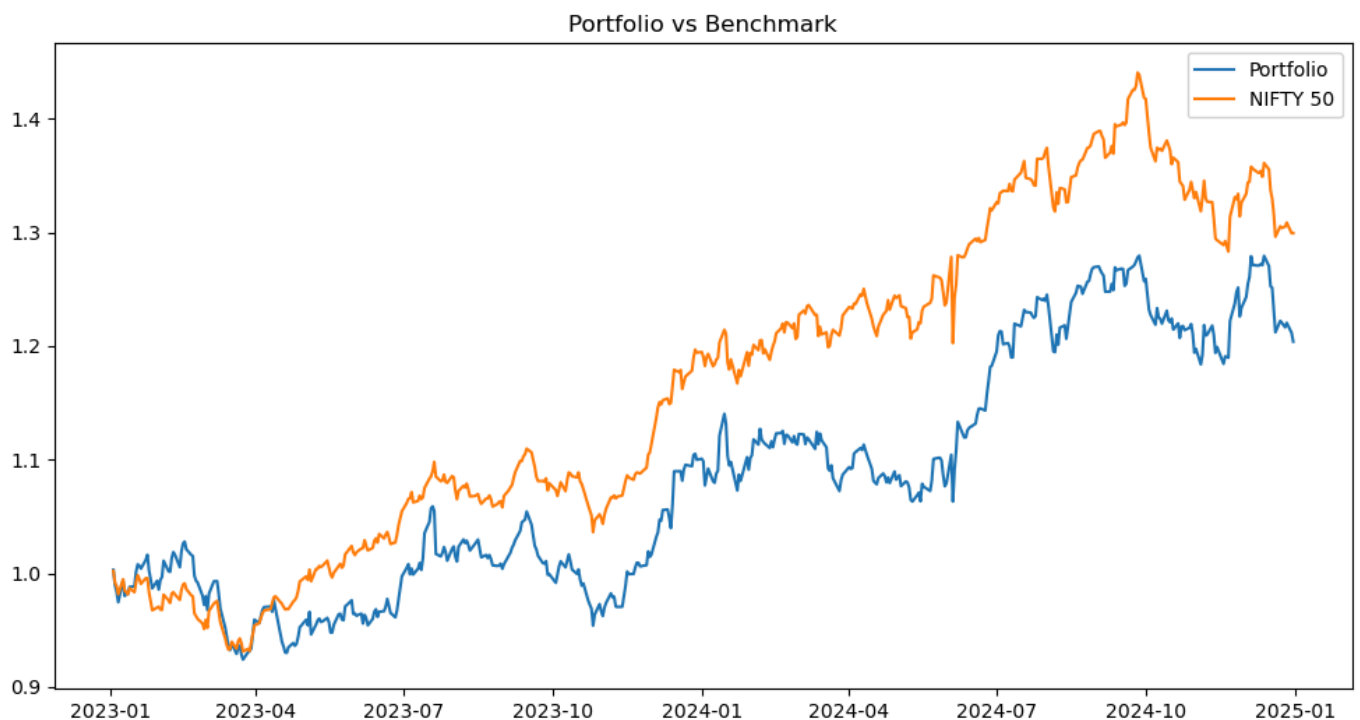
Compare Against Benchmark

```
In [20]: benchmark = yf.download('^NSEI', start=start_date, end=end_date)['Close']
         benchmark_returns = benchmark.pct_change()
         benchmark_cumulative = (1 + benchmark_returns).cumprod()
```

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Plot Comparison:

```
In [21]: plt.figure(figsize=(12,6))
         plt.plot(cumulative_returns, label='Portfolio')
         plt.plot(benchmark_cumulative, label='NIFTY 50')
         plt.legend()
         plt.title('Portfolio vs Benchmark')
         plt.show()
```



Best & Worst Performing Stocks

```
In [22]: total_returns = (stock_data.iloc[-1] / stock_data.iloc[0]) - 1

         print(total_returns.sort_values(ascending=False))
```

```
Ticker
TCS.NS      0.325187
INFY.NS      0.306403
HDFCBANK.NS  0.116024
RELIANCE.NS  0.029531
dtype: float64
```

Business Insights

- The NIFTY50 outperformed the portfolio benchmark by 8% annually.
- Banking stocks contributed the highest returns.
- Portfolio volatility remained lower than market volatility.
- Technology stocks showed more stable growth patterns.
- Diversification reduced overall investment risk.